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Measuring Communicative Intentions and Capabilities: Learners' English Proficiency as a Supplement to Willingness to Communicate*

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In this study, we attempted to measure English learners' communicative intentions and capabilities (CIC) by adding their English proficiency to their willingness to communicate (WTC) as a predictor of the learners' English use. In the context of communicative language teaching (CLT), learners' WTC is considered the key to linking important learner factors such as self-perceived competence, anxiety, and motivation. Given that WTC only accounts for learners' communicative intentions, we hypothesized that learners' communicative capabilities (i.e., language proficiency) would be a meaningful addition to measuring overall CIC. To test this hypothesis, we collected a total of 67 samples (N = 17,811) and conducted an analysis using a meta-analytic structural equation modeling (MASEM) approach. The results indicated that the learners' overall CIC successfully predicted their English use, with a significant path coefficient (β = .525, p = .02). Furthermore, we found that the size of the direct contributions of perceived competence and motivation became non-significantly marginal (ps > .05), indicating that adding learners' proficiency to the WTC to measure the overall CIC can improve the overall explanatory power of the MASEM model to predict learners' English use.

Key words: communicative intentions and capabilities, willingness to communicate, English proficiency, English use, meta-analytic structural equation modeling

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I. INTRODUCTION

1. Background

Willingness to communicate (WTC) is a concept that was originally introduced by McCroskey and his colleagues (e.g., McCroskey, 1992; McCroskey & Baer, 1985; McCroskey & Richmond, 1991) to identify why some are more eager and willing to communicate. In the field of second language (L2) communication, MacIntyre and his colleagues (MacIntyre & Charos, 1996; MacIntyre, Clément, Dörnyei, & Noels, 1998) first applied this concept assuming that WTC would motivate L2 learners to initiate and participate in communication with other L2 speakers in a variety of situations.

According to the communicative language teaching (CLT) approach in L2 learning, which focuses more on L2 use in authentic or real-life communicative situations rather than simply memorizing vocabulary and studying grammar (Khajavy, Ghonsooly, Hosseini, Fatemi, & Choi, 2016), the importance of WTC in L2 has received much scholarly attention because it is significantly related to or directly influences frequent L2 use (Clément, Baker, & MacIntyre, 2003; MacIntyre & Charos, 1996; Yashima, Zenuk-Nishide, & Shimizu, 2004).

WTC in L2 is important not only due to its connection to frequency of L2 use but also for its connection to learners' individual learner factors (e.g., MacIntyre & Charos, 1996). Following MacIntyre and Charos' (1996) conceptual model of L2 WTC, by which L2 WTC is intertwined with other factors and ultimately indicates higher frequency of L2 use, a plethora of studies have explored the model by quantitatively acquiring related empirical evidence or attempting to improve or expand upon the model to investigate the relationships between L2 WTC and learner-individual (especially sociopsychological) variables. For example, L2 motivation (e.g., Ghonsooly, Khajavy, & Asadpour, 2012; Hashimoto, 2002), international posture (e.g., Yashima, 2009; Yashima et al., 2004), perceived L2 competence (e.g., Khajavy et al., 2016; MacIntyre, Baker, Clément, & Donovan, 2003), L2 anxiety (e.g., MacIntyre & Charos, 1996; Papi, 2010), and L2 grit (e.g., Lee & Lee, 2020) have been identified as sociopsychological variables that significantly influence L2 WTC in various social, cultural, and communicative contexts.

Two recent meta-analysis studies (Jin & Lee, 2022; Elahi Shirvan, Khajavy, MacIntyre, & Taherian, 2019) in this area have identified the three high-evidence variables (i.e., self-perceived L2 competence, L2 motivation, L2 anxiety) and investigated their ultimate connection to frequency of L2 use mediated by L2 WTC. In summary, WTC has been confirmed as an important factor that operates based on various sociopsychological and contextual variables and finally affects L2 learners' frequency of communication in their target languages. In particular, Jin and Lee (2022), the most recent meta-analysis on L2 WTC, focused on relevant factors by synthesizing 43 primary studies, and found that L2

WTC is a significant link between learner factors, such as self-perceived L2 competence, L2 anxiety, L2 motivation, and L2 use (see Figure 1).

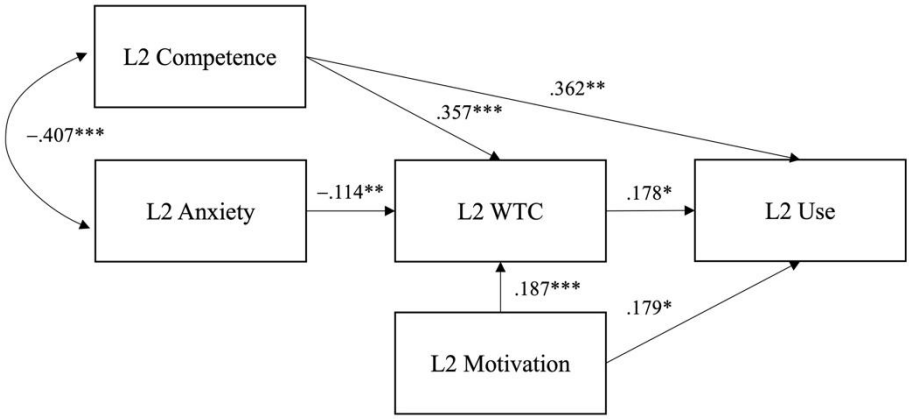


FIGURE 1

Structural Equation Model of Willingness to Communicate (Jin & Lee, 2022)

In the present study, we suggested to reinforce the connection between important learner factors and the frequency of L2 use, based on an evidence-based relation between learners' L2 WTC and L2 proficiency (e.g., Alemi, Daftarifard, & Pashmforoosh, 2011; Darasawang & Reinders, 2021; Yashima, 2009). Specifically, L2 learners' WTC in L2 is a measure of their communicative *intentions*. In other words, as MacIntyre, Clément, Dörnyei and Noels (1998) eloquently worded it, WTC for L2 communication represents one's readiness or intention to communicate with a variety of people using a target language. Despite its close relationship with L2 WTC, the importance and role of L2 proficiency in the context of CLT have not been seriously viewed, as this construct has not been considered as an ultimate goal of L2 learning (e.g., Yashima et al., 2004). In the present study, we argue that L2 proficiency is an important measurement for representing one's communicative *capabilities*, which are on par with one's communicative intentions in terms of importance of linking learner factors and L2 use. Therefore, we conjecture that considering both communicative intentions and capabilities (CIC) is a more accurate and reliable way to predict L2 learners' communicative behaviors (i.e., frequency of L2 use).

2. Literature Review

The main focus of communicative language teaching (CLT) is to develop the structural and functional aspects of communication in L2 learners' target language (Littlewood, 1981) by using the language both meaningfully and effectively in a variety of real-life or authentic

communicative contexts (Johnson, 1982; Johnson & Brumfit, 1979; Larsen-Freeman, 2000; Littlewood, 1981; Savignon, 1991). In this sense, WTC plays a crucial role in the CLT approach as an essential factor that influences L2 learning, as this construct represents learners' readiness or preparedness to initiate interaction and communication with other L2 speakers (MacIntyre & Charos, 1996).

Unlike WTC, which has been acknowledged as playing a critical role in L2 communication from the CLT perspective, learners' L2 proficiency has not received much attention, as it is not considered as "the goal of learning an L2 per se" (Yashima et al., 2004, p. 120). From the CLT perspective, as the name indicates, the main goal of L2 learning is learners' active use of their target language (Larsen-Freeman, 2000). Because the CLT approach places more importance on L2 communication itself rather than structural knowledge (e.g., vocabulary and grammar), the learner's intention (i.e., WTC) to participate in effective L2 communication is considered the most crucial aspect of L2 learning.

Nevertheless, there has been a continuous effort to investigate the role of L2 proficiency in the context of the CLT perspective (e.g., Alemi et al., 2011; Darasawang & Reinders, 2021; Edwards & Roger, 2015; Lodhi, Yousaf, Hassan, Anwar, & Farman, 2019; Rostami, Kashanian, & Gholami, 2016; Tan & Phairot, 2018). These studies were conducted by analyzing the relationship between L2 proficiency and L2 WTC, as these two constructs are believed to perform similar functions in the context of the CLT approach, assuming that one's actual capabilities in communication (i.e., proficiency) may be closely related with one's intention to communicate (i.e., WTC).

For example, in a study of 49 Iranian university students, Alemi et al. (2011) found that high proficiency learners had higher WTC. Based on 60 high school and university students' data, Rostami et al. (2016) confirmed a significant relationship between the two constructs such that high-proficiency learners were more willing to communicate. Likewise, with 375 high school students, Tan and Phairot (2018) confirmed a significant relationship between L2 proficiency and L2 WTC. Last but not least, Darasawang and Reinders (2021) confirmed a statistically significant correlation between the two variables ($r = .26$) with a study of 1,836 university students.

Then, given its close relationship with WTC, the role of proficiency in CLT may be closely related to the improvement of "functional language ability" (Savignon, 1991, p. 265) by means of active participation in authentic L2 communication. Proficiency in CLT, therefore, refers not only to the mastery of vocabulary knowledge and grammatical rules but also to the capability to utilize them in various communicative situations and negotiate meanings with others. In other words, along with WTC, learners' L2 proficiency is believed to enable learners to adapt their target language use to various sociocultural contexts in order to achieve their communicative objectives, thereby contributing to the ultimate goal of frequent L2 communication in authentic communication situations.

Although previous studies continuously investigated how learners' L2 proficiency is related with their L2 WTC, it is still not clear what roles L2 proficiency plays in the context of L2 WTC which connects important learner factors and the frequency of L2 use. Therefore, in the present study, we argue that if we consider the importance of L2 learners' WTC and proficiency as inseparable constructs given their close relationship, it is implied that these two constructs should both be taken into account when predicting learners' frequent use of the L2 (i.e., L2 communication behaviors). For this purpose, we suggest the concept of CIC, which combines these constructs to better predict learners' frequent use of their target language from the CLT perspective.

3. Present Study

In the present study, we employed a meta-analytic structural equation modeling (MASEM) approach to measure CIC as a latent variable consisting of both willingness to communicate and proficiency. To ensure consistency with the latest MASEM model, the attempt was made in the context of CLT, in which one's willingness to communicate connects learner factors such as self-perceived L2 competence, L2 anxiety, and L2 motivation, as shown in Figure 2.

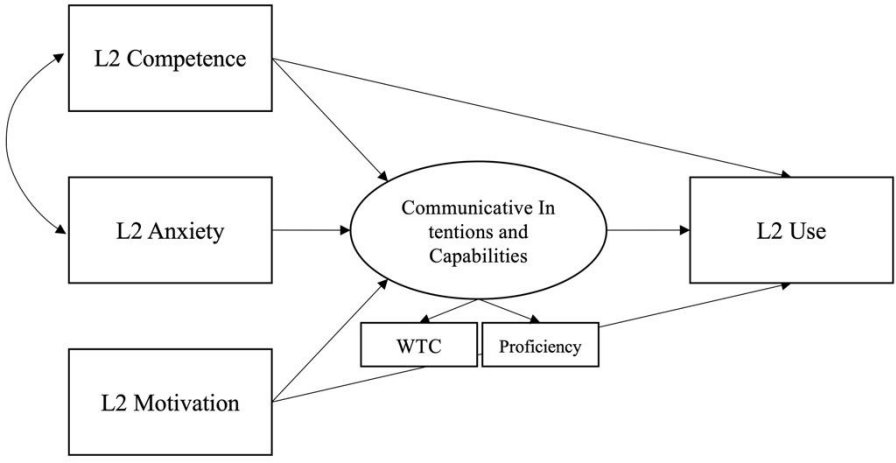


FIGURE 2

Proposed Structural Equation Model with Communicative Intentions and Capabilities

The MASEM approach is to build a structural equation modeling (SEM) model by fitting a pooled correlation matrix, which can be produced by averaging (with different weights) a set of correlation coefficients across variables of interest collected from primary empirical studies. Thus, we can investigate the structural relationships between multiple variables of interest (i.e., a multivariate model; Kline, 2005) from a systematic and comprehensive

perspective based on aggregated data from objectively collected literature (i.e., meta-analysis, e.g., Lee, Warschauer, & Lee, 2019). The following research questions have guided this study.

Research question #1. Does the collected data set successfully fit the proposed MASEM model with CIC as a latent variable?

Research question #2. How does the CIC measurement change the overall relationships between the variables of interest compared to the latest MASEM model?

II. RESEARCH METHOD

1. Data Collection

To collect data for the current meta-analysis, we conducted database and journal searches using the following combination of keywords: "willingness to communicate" AND ["second language" OR "foreign language"] AND ["motivation" OR "anxiety" OR "competence" OR "proficiency"]. For databases, we used Web of Science, SCOPUS, and ProQuest. For journals, we browsed relevant scholarly journals with high impact factors such as *Annual Review of Applied Linguistics, Applied Linguistics, ELT Journal*, *Language Learning, Language Teaching Research, Studies in Second Language Acquisition, Studies in Second Language Learning and Teaching, System, TESOL Quarterly,* and *The Modern Language Journal*. Once an article was identified, its reference list was reviewed to further search for related literature. To be included in the current meta-analysis, a study was required to (1) be written in English, (2) be conducted in language learning contexts, and (3) report correlation coefficients between our variables of interest along with sample size.

As we identified suitable studies from these resources, we reviewed their titles and abstracts to check their eligibility for a full review of their contents, resulting in a total of 131 eligible studies remaining for the next step. After the full reviews, a total of 66 studies were excluded for the following reasons: (1) 20 studies were not related to WTC, (2) 34 studies did not report raw correlation coefficients, and (3) 12 studies did not have our variables of interest in their datasets. Finally, a total of 65 studies were considered for this meta-analysis. Among them, Ghonsooly et al. (2012) and MacIntyre et al. (2003) each had two independent samples in their studies; therefore, there were a total of 67 samples (N = 17,811) across these studies. It should be noted here that as the current study was conducted as a follow-up study to Jin and Lee (2022), in which five out of our six variables of interest were considered for the MASEM model. The dataset built for the current study covered all samples (k = 44, n = 12,094) previously included in Jin and Lee (2022), indicating that we added a total of 23 new samples (n = 5,717) in this study.

2. Dataset Construction

The 67 samples identified were further reviewed to code the correlation coefficients along with the sample sizes. As there we had six variables of interest, namely, learners' perceived competence, anxiety, motivation, willingness to communicate, proficiency, and English use, we coded any of the 15 types of correlation coefficients among the variables. When defining the variables' measurement methods, we followed those used in Jin and Lee (2022). For example, to measure learners' WTC, the most common instrument was McCroskey's (1992) WTC scale, followed by two variations developed by MacIntyer, Baker, Clément and Conrod (2001) and Weaver (2005). Similarly, for measuring learners' motivation, anxiety, and perceived competence, there were a variety of instruments used (e.g., Gardner's [1985] socio-educational motivation model, Deci and Ryan's [1985] self-determination theory, and Dörnyei's [2009] L2 motivational self-system for the motivational aspect; Horwitz's [1986] foreign language classroom anxiety scale, McCroskey, Richmond and McCroskey's [1987] communication apprehension scale for the anxiety aspect; and McCroskey et al.'s [1987] communicative competence scale for the perceived competence aspect) to measure the same constructs. With respect to learners' proficiency, which was not considered by Jin and Lee (2022), we found that the majority of the included studies attempted to measure learners' overall communicative capabilities by employing either researcher-developed or standardized tests. Each study reported correlation coefficients for each of our variables of interest, which we coded into the dataset. Following Jin and Lee (2022), we averaged the correlation coefficients when multiple coefficients for the same pairs of variables were reported in a single study. The values in Table 1 indicate the number of raw correlation coefficients collected from the included 67 samples along with the total sample sizes for each pair of target variables.

TABLE 1

Number of Raw Correlation Coefficients and Sample Sizes

	Var 1	Var 2	Var 3	Var 4	Var 5	Var 6
Var 1: L2 WTC	-					
Var 2: L2 Motivation	30 (9,038)	-				
Var 3: L2 Anxiety	31 (9,776)	36 (9,120)	-			
Var 4: L2 Competence	27 (8,709)	25 (6,618)	28 (8,874)	-		
Var 5: L2 Use	8 (1,798)	9 (1,726)	6 (1,409)	5 (1,271)	-	
Var 6: L2 Proficiency	12 (5,460)	13 (3,307)	18 (4,991)	10 (4,770)	2 (1,990)	-

Note. Values indicate the number of raw correlation coefficients for the pairs of target variables; values in parentheses indicate the total sample size of the related samples for the corresponding pairs of target

variables.

3. Two-Stage Structural Equation Modeling (TSSEM)

To conduct the MASEM, we employed the two-stage structural equation modeling (TSSEM) method developed by Cheung and Chan (2005). To this end, we used "metaSEM: An R Package for Meta-Analysis using Structural Equation Modeling" version 1.3.0 (Cheung, 2015) in R version 4.2.2.

1) First Stage: Computing a Pooled Correlation Matrix

In the first stage, the TSSEM package enabled us to build a pooled correlation coefficient matrix by averaging raw correlation coefficients collected from the included 67 samples. As each sample had different sizes, weighted averages accounting for the importance of each raw correlation coefficient according to the original sample size were computed.

2) Second Stage: Building a Structural Equation Model

In the second stage, the TSSEM package was used to fit the SEM model based on the pooled correlation coefficient matrix constructed in the first stage. As a regular SEM model, we use five model fit indices (Kline, 2005) to check if the dataset successfully fit the proposed SEM model, such as (1) chi-square test (either $p > .05$ or $\chi^2/df < 5$), (2) root mean square error of approximation (< .08), (3) standardized root mean square residual (< .08), (4) comparative fit index (> .90), and (5) Tucker–Lewis index (> .95). Jin and Lee (2022), Lee, Warschauer and Lee (2020), and Lee, Jung and Lee (2022) provide practical examples and guidelines.

III. RESULTS AND DISCUSSION

1. Pooled Correlation Coefficient Matrix

The values in Table 2 indicate the pooled correlation coefficients for pairs of target variables, with the values in parentheses indicating the standard errors of the pooled correlation coefficients. All pooled correlation coefficients were statistically significant (*p*s < .05).

TABLE 2

Pooled Correlation Coefficient Matrix

	Var 1	Var 2	Var 3	Var 4	Var 5	Var 6
Var 1: L2 WTC	-					
Var 2: L2 Motivation	0.33 (0.02)	-				
Var 3: L2 Anxiety	-0.32 (0.03)	-0.26 (0.03)	-			
Var 4: L2 Competence	0.47 (0.03)	0.35 (0.03)	-0.39 (0.03)	-		
Var 5: L2 Use	0.39 (0.05)	0.34 (0.05)	-0.30 (0.04)	0.43 (0.09)	-	
Var 6: L2 Proficiency	0.25 (0.03)	0.26 (0.05)	-0.31 (0.03)	0.33 (0.05)	0.15 (0.02)	-

Note. Values indicate the pooled correlation coefficients for pairs of target variables based on the collected raw correlation coefficients; values in parentheses indicate the standard errors of the corresponding pooled correlation coefficients. All pooled correlation coefficients were statistically significant ($ps < .05$).

When compared with the pooled correlation coefficient matrix reported by Jin and Lee (2022), which included all except the proficiency variable in the data analysis, we found that the 10 pooled correlation coefficients across the five variables (i.e., Variables 1 to 5) were almost identical. With respect to the remaining five pooled correlation coefficients between L2 proficiency and the other variables, we found that L2 proficiency variable had small but significant correlations with the other five. In particular, the results indicated that the overall correlation between L2 WTC and L2 proficiency based on 12 raw correlation coefficients from 12 samples (n = 5,460) was statistically significant ($p < .001$) and its overall average size was .25 (SE = .03). This finding corresponds to previous empirical studies in which a close relationship between L2 proficiency and L2 WTC was continuously reported (e.g., Alemi et al., 2011; Darasawang & Reinders, 2021; Edwards & Roger, 2015; Lodhi et al., 2019; Rostami et al., 2016; Tan & Phairot, 2018). While some studies (e.g., Yashima et al., 2004) showed no significant correlation between L2 proficiency and L2 WTC, our research synthesis of relevant previous studies revealed that the relationship was overall found to be positive and statistically significant. In the next section, an SEM model is built and tested, and a more accurate evaluation of the role of L2 proficiency in the context of CLT is conducted based on the significant correlation between L2 proficiency and L2 WTC.

2. Meta-Analytic Structural Equation Model

In the second stage, we built an SEM model based on the pooled correlation coefficient matrix (Table 2) computed in the first stage. Figure 3 shows the results of our tested SEM model, including a latent variable for CIC, composed of learners' L2 WTC and L2 proficiency. According to the model fit indices, we found that our tested SEM model based on a total of 67 samples (N = 17,811) had an overall acceptable model fit, with (1) $\chi^2(4)$ = 11.61, p = .02, χ^2/df = 2.9; (2) CFI = .98; (3) TLI = .94; (4) RMSEA = .01; (5) SRMR = .03.

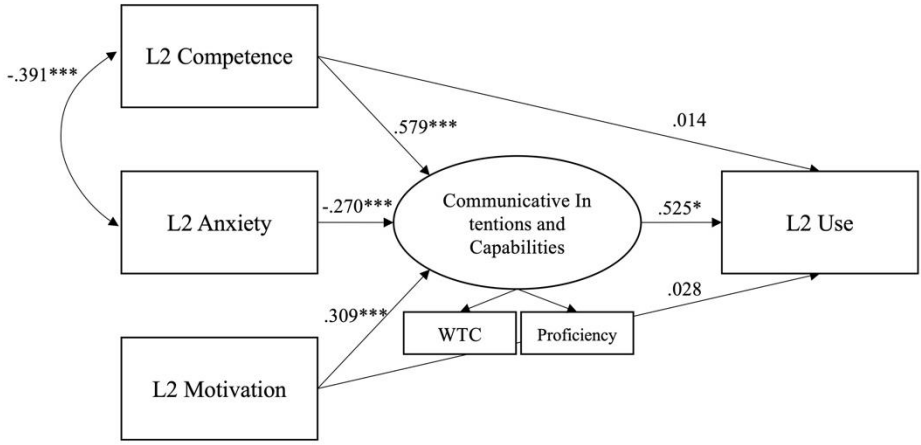


FIGURE 3

Structural Equation Model with Communicative Intentions and Capabilities

The latent variable for CIC, which was composed of learners' L2 WTC and L2 proficiency, showed a similar but substantially greater role in linking learner factors and L2 use, compared to the role of L2 WTC by itself in Jin and Lee (2022). For example, the path coefficient of L2 WTC toward L2 use was .178 in Jin and Lee's (2022) study, whereas that of CIC was .525 in the present study. In addition, the contributions of learner factors to the linkage were greater, such that L2 competence had a direct contribution of .579, L2 anxiety had -.270, and L2 motivation had .309, whereas the path coefficients of these variables were .357, -.114, and .187, respectively, in Jin and Lee (2022). Although we cannot statistically compare the two sets of results from the two independent models, we can clearly observe that the absolute values of their path coefficients are substantially different.

Furthermore, our inclusion of L2 proficiency to measure learners' CIC contributed to a remarkable change: the direct impacts of learners' self-perceived L2 competence and L2 motivation on their frequent L2 use were marginal and not statistically significant (ps > .05). One possible reason is that the inclusion of L2 proficiency along with L2 WTC as the linkage between learner factors and L2 use greatly contributed to enhancing the predictive power of the newly built latent variable to better explain learners' frequent L2 use, by absorbing the direct contributions of L2 competence and L2 motivation (e.g., βs = .362 and .179 [both were

statistically significant; *p*s < .05], respectively, in Jin & Lee, 2022) that otherwise may have existed in the absence of CIC. Taken together, we could argue that learners' L2 proficiency matters even in the context of CLT and plays a significant role in learners' communicative intentions and capabilities (particularly the latter aspect) to directly influence the frequency of learners' target language use. In this sense, learners' L2 proficiency, while not the ultimate goal of L2 learning (e.g., Yashima et al., 2004), needs to be seriously considered a learning objective along with WTC and other important learner factors (e.g., MacIntyre & Charos, 1996) to improve learners' active L2 use.

IV. CONCLUSION

In the present study, we suggested that measuring learners' CIC (communicative intentions and capabilities) by combining their WTC and proficiency as the latent variable connecting important learner factors with the frequency of L2 use may enhance their explanatory power to predict learners' frequency of L2 use. To this end, we adopted a MASEM (meta-analytic structural equation modeling) approach, which is a combination of meta-analysis and structural equation modeling and collected a total of 67 samples (N = 17,811) to build a pooled correlation coefficient matrix for a proposed MASEM model with a latent variable to measure learners' CIC. The results of our MASEM approach confirmed that learners' CIC, composed of L2 WTC and L2 proficiency, played an important role in mediating the impact of important learner factors on their frequency of L2 use. That is, the newly built CIC variable had (arithmetically, not statistically) greater path coefficients not only from learner factors to the CIC but also from the CIC to the frequency of L2 use when compared to the results of the previous MASEM approach conducted by Jin and Lee (2022).

This study has both methodological and pedagogical implications. Regarding methodology, our use of the MASEM approach is unique in the field of English assessment in Korea. This method includes the strengths of both meta-analysis and structural equation modeling, and can be useful to those who aim to investigate simultaneous associations among multiple variables in a structured relationship based on multiple samples collected from previously published works (Cheung & Chan, 2005). As shown in the present study, one can investigate the role of a variable in the context of multiple variables by collecting and coding related empirical studies with greater statistical power, accuracy, reliability, and generalizability than a single empirical field study with a small number of participants.

Regarding pedagogical implications, the results of the present study highlight the importance of considering learners' actual communication capabilities when focusing on enhancing their communicative intentions. We found that learners may require a sufficient level of proficiency to engage in L2 communication, assuming that they have a sufficient

level of L2 WTC. Among the important learner factors, the literature has focused on learners' self-perceived L2 competence in their L2 communication (e.g., e.g., Khajavy et al., 2016; MacIntyre & Charos, 1996; MacIntyre et al., 2003), and the present study revealed that not only learners' beliefs towards their communicative capabilities, but also their actual capabilities matter in the context of improving the frequency of their L2 use. Therefore, if educators and language teachers could reliably assess learners' CIC, they could more efficiently provide appropriate learning opportunities for learners' active participation in L2 communication.

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APPENDIX

List of the Included 63 Studies

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Applicable levels: Secondary and Tertiary

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